Ruthvik Varma Dendukuri

Data Science and AI undergraduate specializing in software development, artificial intelligence, and machine learning.

9885801818 ruthvikvarma24@gmail.com LinkedIn Portfolio Hyderabad, India

PROFESSIONAL SUMMARY

Data Science and AI undergraduate with hands-on experience in **software application development**, **Python programming**, **SQL**, **REST API development**, and **AI-powered applications**. Experienced in **designing, developing, integrating, and validating software solutions** through academic projects and internships. Passionate about **building scalable applications**, **solving real-world problems**, and continuously learning new technologies.

WORK EXPERIENCES

AI/ML Intern | Internship — Jan 2026 - Aug 2026
NextEra Unitech Pvt Ltd — *On-site • Hyderabad, Telangana*

- Collected, cleaned, organized, and analyzed structured datasets to support **software application development**, data preprocessing, model training, and performance evaluation.
- Developed and optimized **software applications** and computer vision models using **TensorFlow, YOLOv8, and MediaPipe**, achieving **90–95% accuracy and recall**.
- Developed, integrated, and validated **REST APIs** and software components to automate data processing and improve application reliability.

Technologies / Skills Used : *TensorFlow, YOLOv8, MediaPipe, Google Sheets, SQL*

SEO Analyst | Internship — Jun 2024 - Aug 2024
Medintu Health Solutions — *On-site • Hyderabad, Telangana*

- Improved organic search rankings to **top-5 positions** through **on-page and off-page SEO strategies**.
- Increased organic traffic by **30%** via **keyword analysis** and **content optimization**.

Technologies / Skills Used : *SEO, Keyword Research, SemRush, Ahref*

PROJECTS

AI Pharmacovigilance & Medication Safety Platform

Developer

- Developed a full-stack **software application** for pharmacovigilance to analyze patient medication records and improve medication safety.
- Built deterministic analytics engines to detect drug-drug interactions, adverse drug reactions (ADRs), medication adherence, and patient safety risks.
- Designed SQL-based evidence retrieval and explainable safety scoring pipelines with complete audit trails for clinical decision support.
- Developed secure **REST APIs**, implemented **JWT authentication**, and designed normalized **PostgreSQL** schemas to manage longitudinal patient health records.

Technologies / Tools Used : *Python, SQL, PostgreSQL, FastAPI, SQLAlchemy, Supabase, LangGraph, REST APIs, JWT Authentication, Docker, Git*

Real-Time Attention Monitoring System

Developer

- Built a **real-time software application** for attention monitoring, classifying users as Attentive, Distracted, Sleepy, or Tired at **30+ FPS**.
- Detected fatigue and yawning using EAR/MAR metrics with MediaPipe's 468-point facial landmark model.
- Implemented gaze tracking to identify distraction with **90%+ accuracy**.
- Reduced redundant detections by **60%** using **YOLOv8 segmentation** combined with PCA and cosine similarity.

Technologies / Tools Used : *Python, MediaPipe, YOLOv8, OpenCV*

Squint Eye (Strabismus) Detection

Developer

- Built a **MobileNetV2-based deep learning model** detecting strabismus with **92%+ accuracy**.
- Improved model precision by extracting eye regions using MediaPipe facial landmarks.
- Developed a **Streamlit web application** delivering predictions in **under 2 seconds**.
- Ensured robustness via fallback to full-image analysis when landmark detection fails.

Technologies / Tools Used : *Python, MobileNetV2, TensorFlow, MediaPipe, Streamlit*

Microhabit Deviation System with AI Companion

Developer

- Built an **ML-driven system** to track microhabits and detect behavioral deviations using pattern analysis on user activity logs.
- Developed an **AI companion using NLP** to generate personalized recommendations, reminders, and corrective strategies.
- Implemented deviation detection logic based on historical trends to identify inconsistencies such as missed tasks and declining performance.
- Built a **full-stack software application** with web interfaces, backend APIs, and persistent data storage to support long-term habit tracking.

Technologies / Tools Used : *Python, ML, NLP, Full-Stack*

Colorization of Images

Developer

- Developed a **deep learning model** for **automatic image colorization**, enhancing generalization through preprocessing and augmentation techniques.
- Refined model architecture and hyperparameters by evaluating outputs against ground-truth images, resulting in more **realistic color predictions**.

Technologies / Tools Used : *Python, Deep Learning, OpenCV*

SKILLS

Programming Languages :	Python, SQL, HTML, CSS
Frameworks & Libraries :	FastAPI, TensorFlow, OpenCV, MediaPipe, YoloV8, SQLAlchemy, Pandas, Numpy, Scikit-learn, Streamlit
Databases :	PostgreSQL, MySQL, Supabase
Tools & Platforms :	VS Code, Google Colab, Jupyter Notebook, Kaggle Notebook, Git, Streamlit, Google Workspace, Docker
Soft Skills :	Problem Solving, Critical Thinking, Time Management, Teamwork, Communication, Adaptability
Languages :	English, Hindi, Telugu

EDUCATIONS

ICFAI Foundation for Higher Education | Donthanpalle, Telangana — 2022 - 2026
Bachelor of Technology in Data Science and AI

Narayana IIT Academy | Kondapur, Telangana — 2020 - 2022
Board of Intermediate Education (MPC)

Silver Oaks International School | Bachupally, Hyderabad — 2019 - 2020
Central Board of Secondary Education

CERTIFICATIONS

Infosys Springboard: Data Analysis with Python
Kaggle: Intro to Programming | Python
NASSCOM Future Skills: Acquiring Data | Exploratory Data Analysis | Data Processing and Visualization

EXTRA CURRICULAR

- Earned the **NCC 'A' Certificate**; participated in camps and developed **leadership, teamwork, and discipline** through parades and drills.